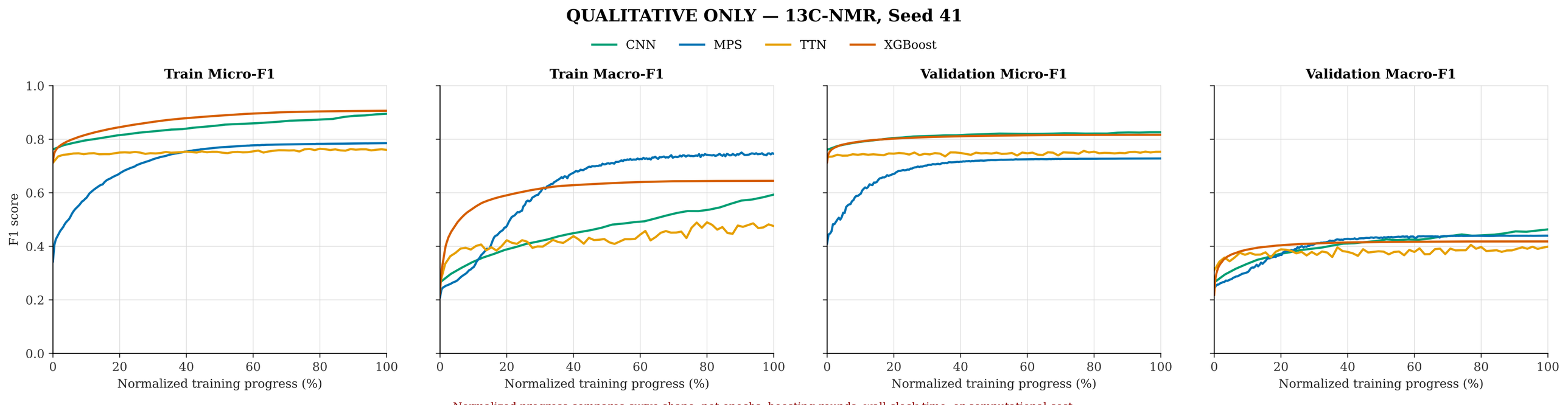
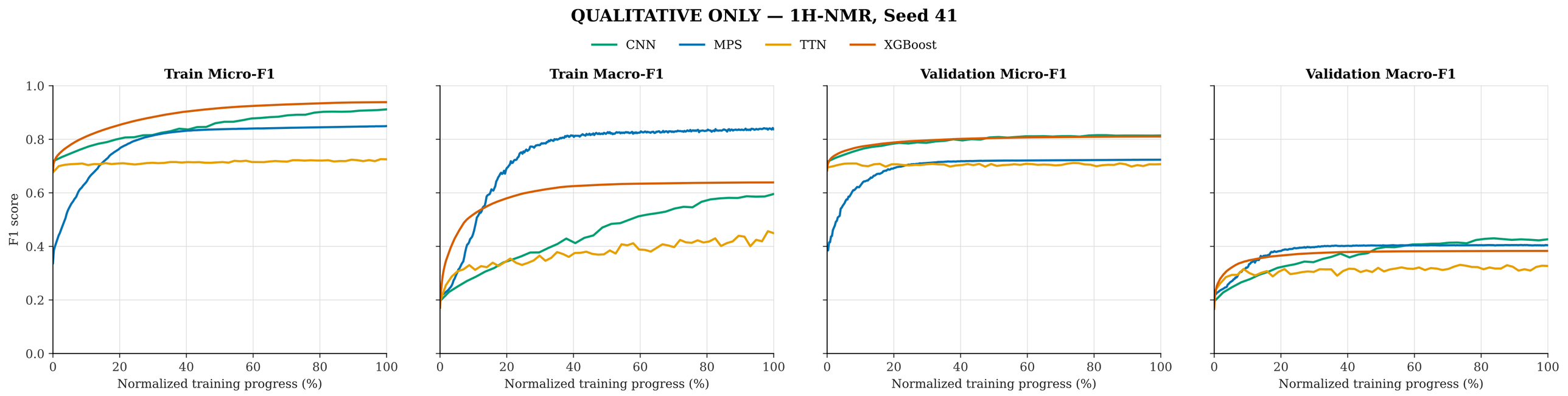
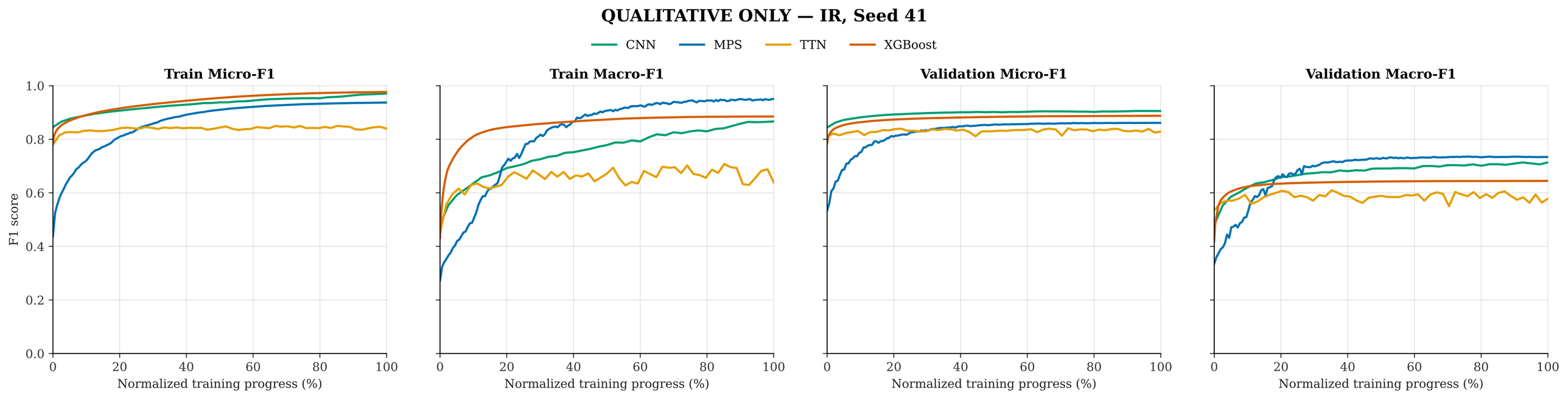
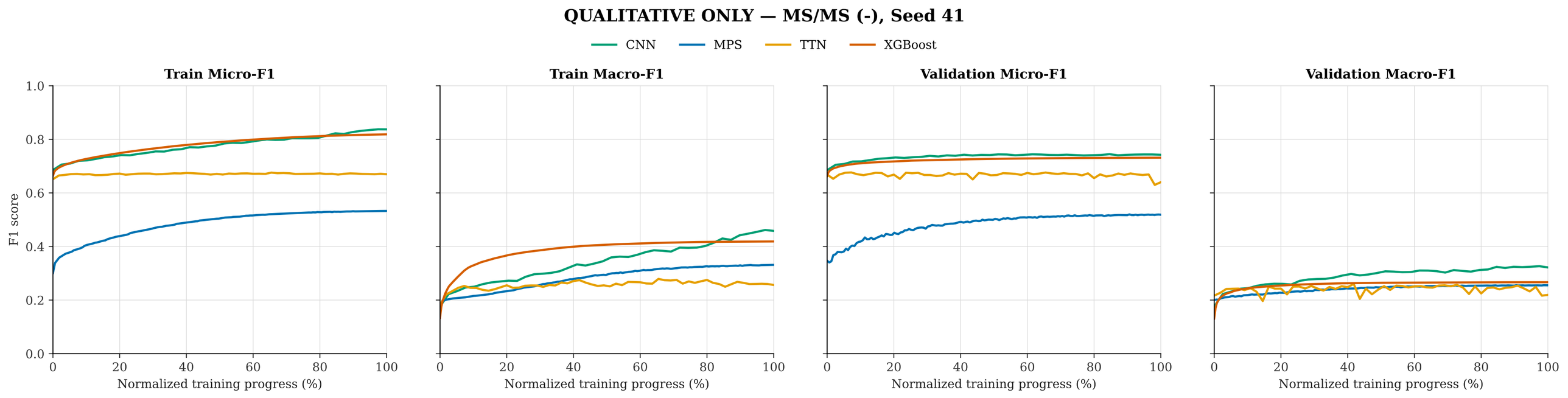
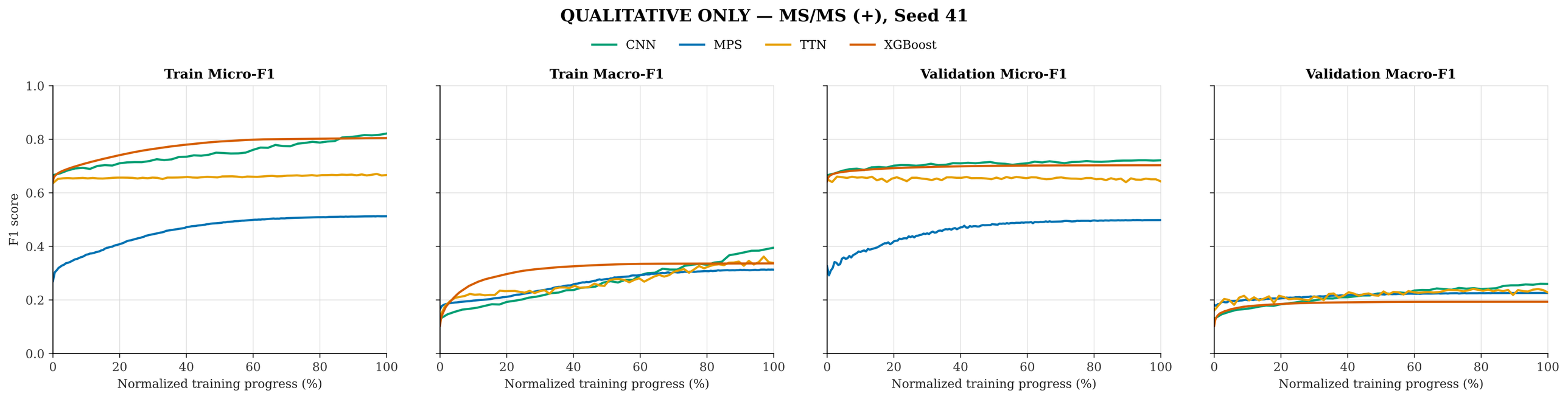
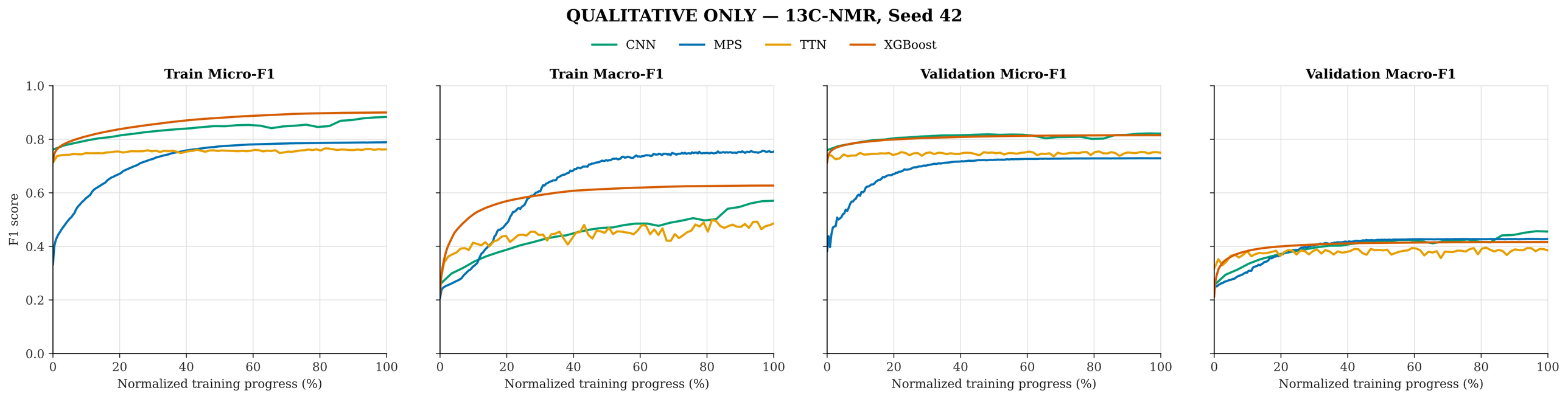

QUALITATIVE ONLY — 13C-NMR, Seed 41
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

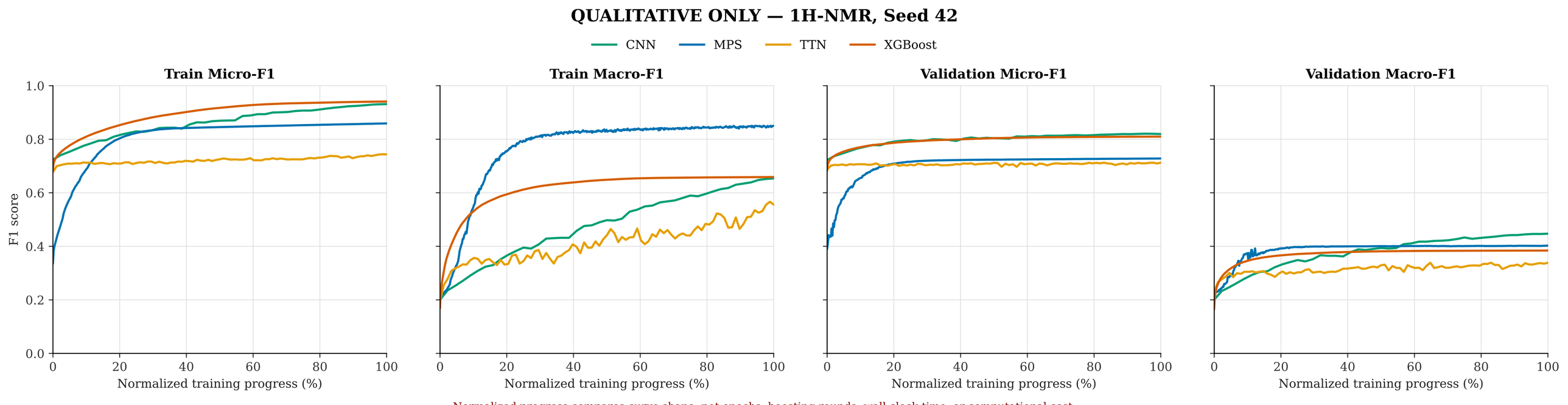

QUALITATIVE ONLY — 1H-NMR, Seed 41
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

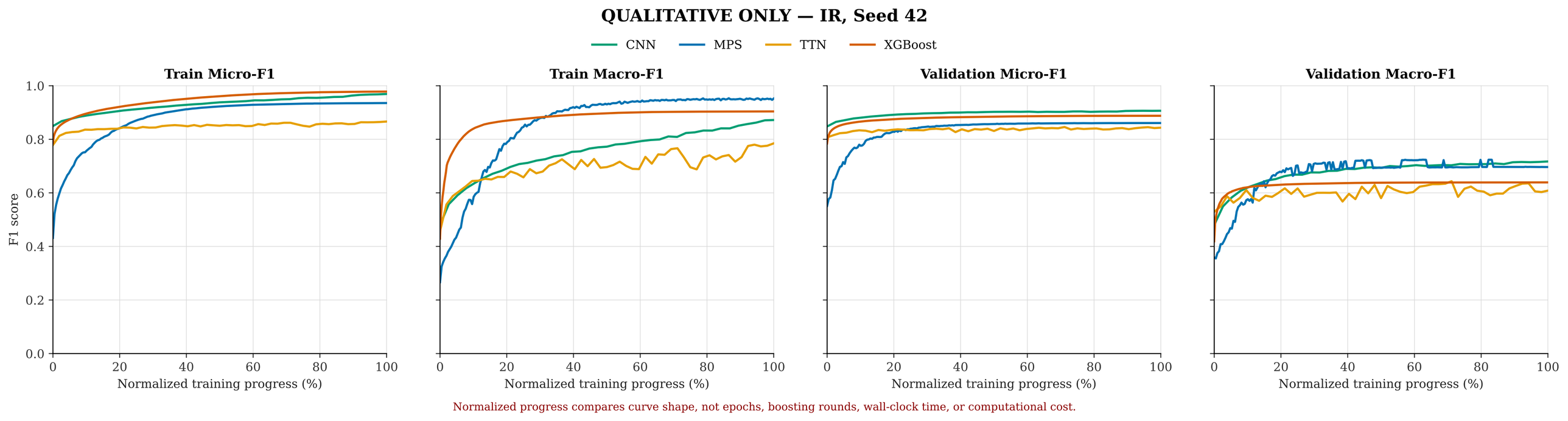

QUALITATIVE ONLY — IR, Seed 41
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

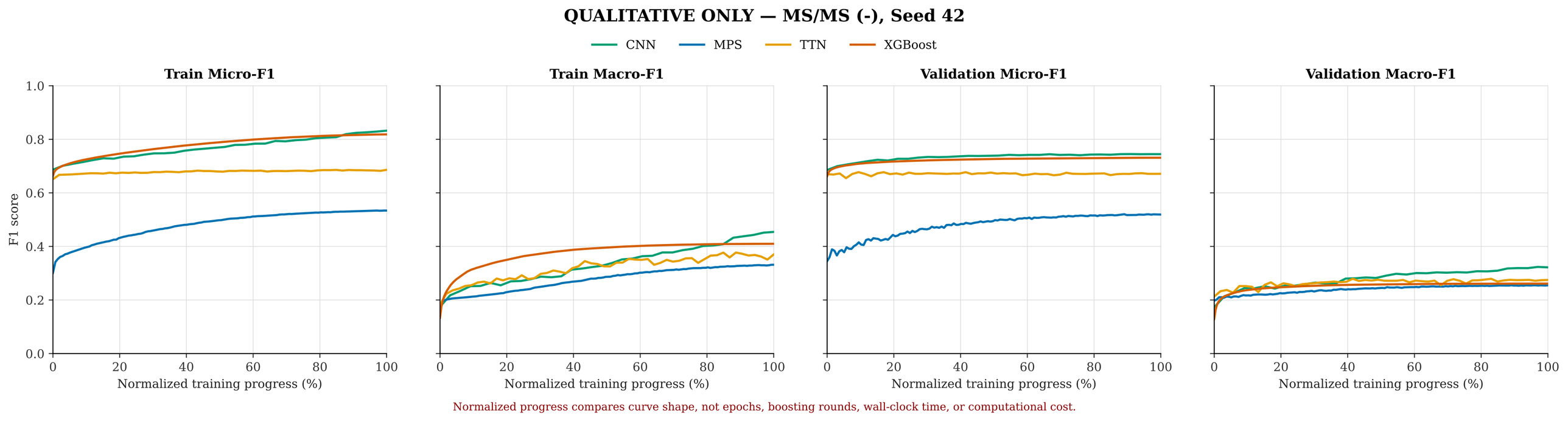

QUALITATIVE ONLY — MS/MS (-), Seed 41
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

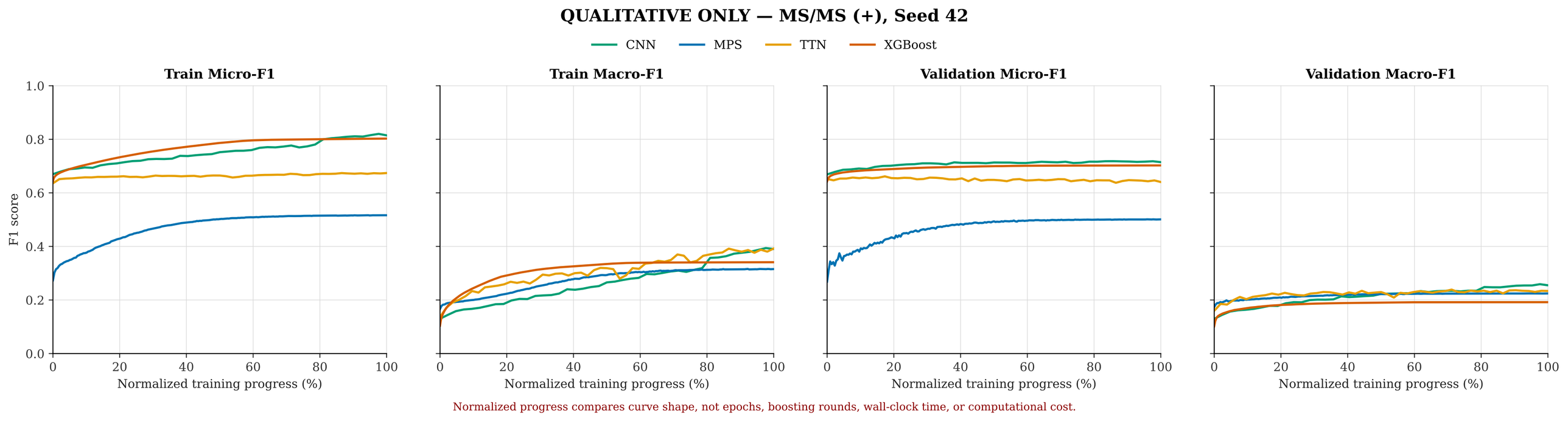

QUALITATIVE ONLY — MS/MS (+), Seed 41
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

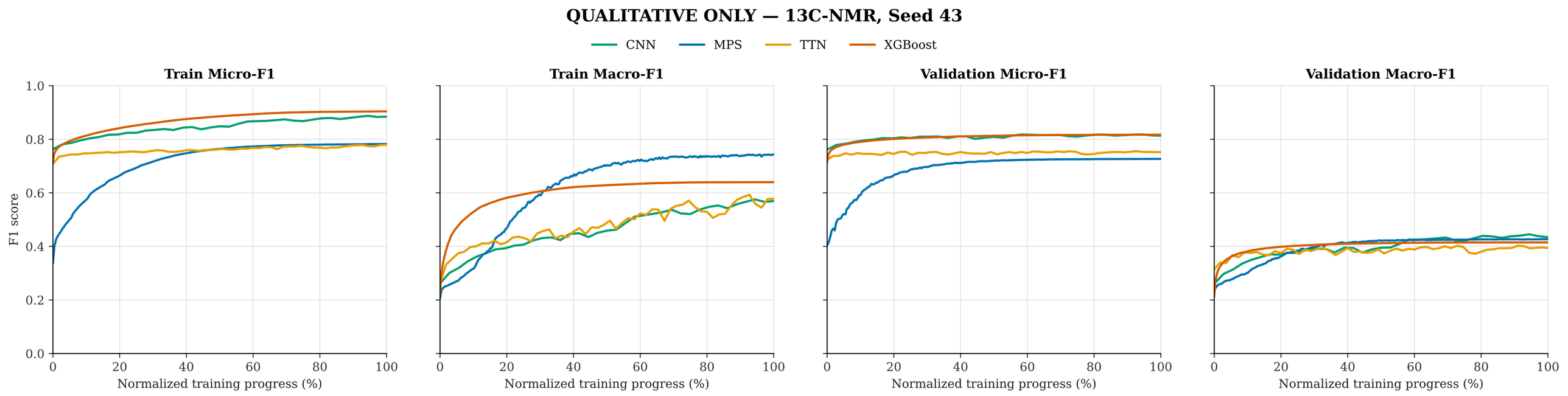

QUALITATIVE ONLY — 13C-NMR, Seed 42
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

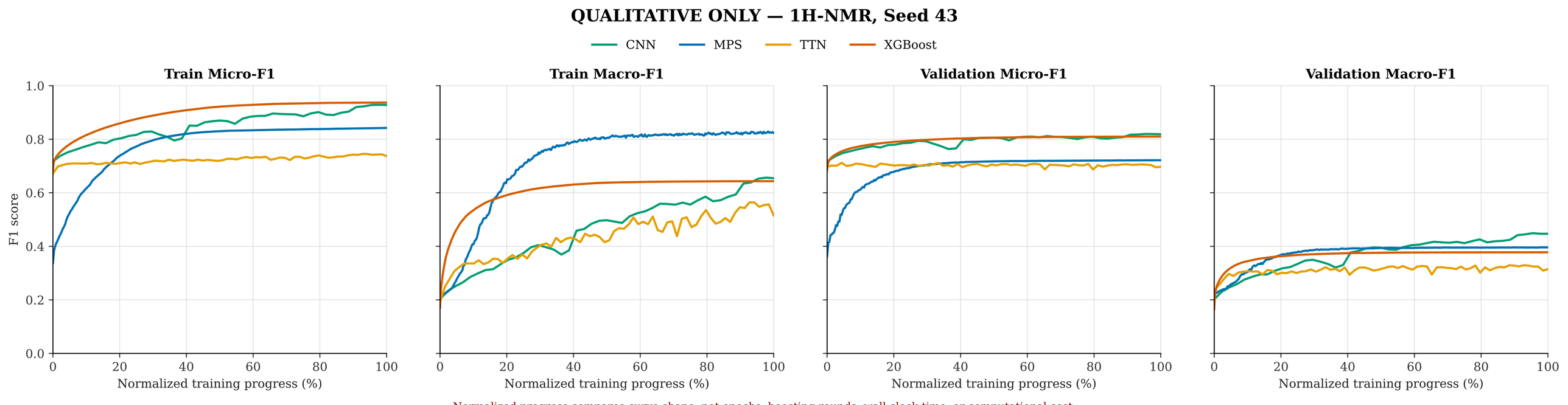

QUALITATIVE ONLY — 1H-NMR, Seed 42
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

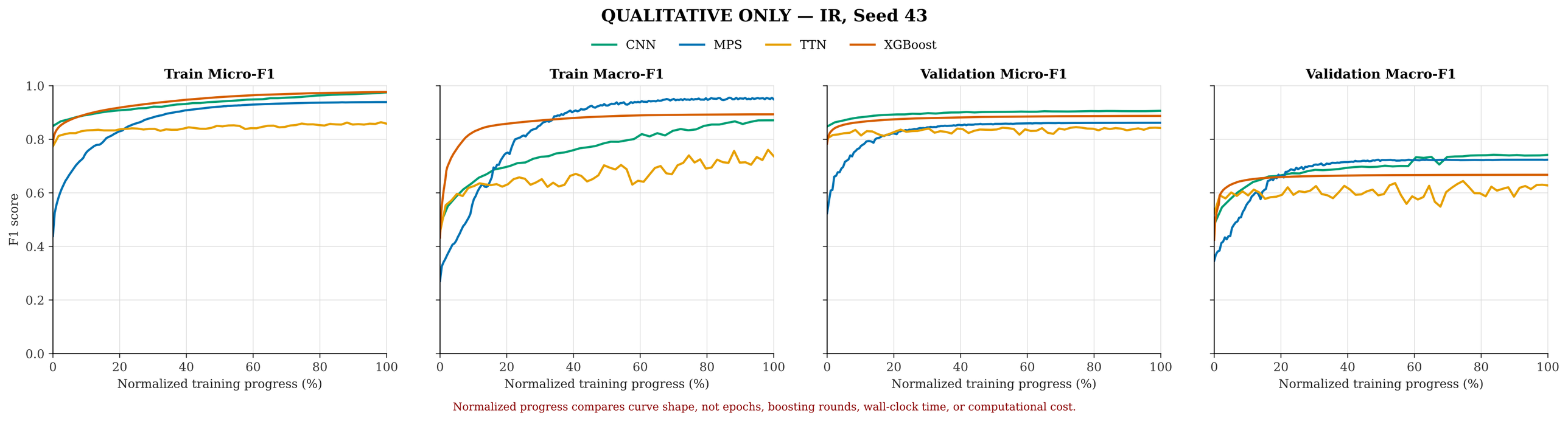

QUALITATIVE ONLY — IR, Seed 42
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

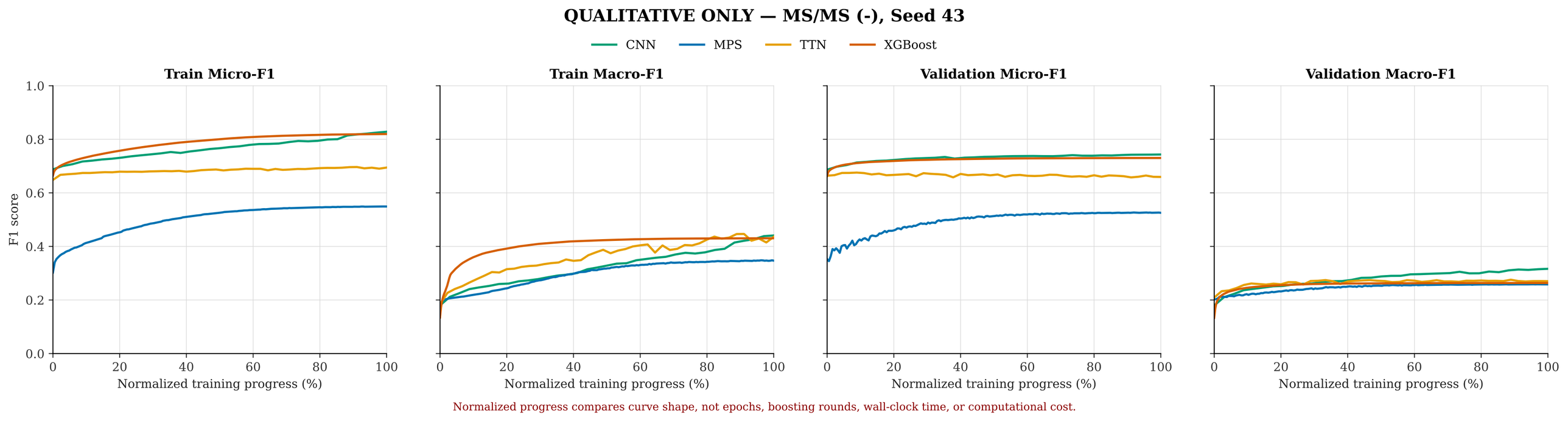

QUALITATIVE ONLY — MS/MS (-), Seed 42
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

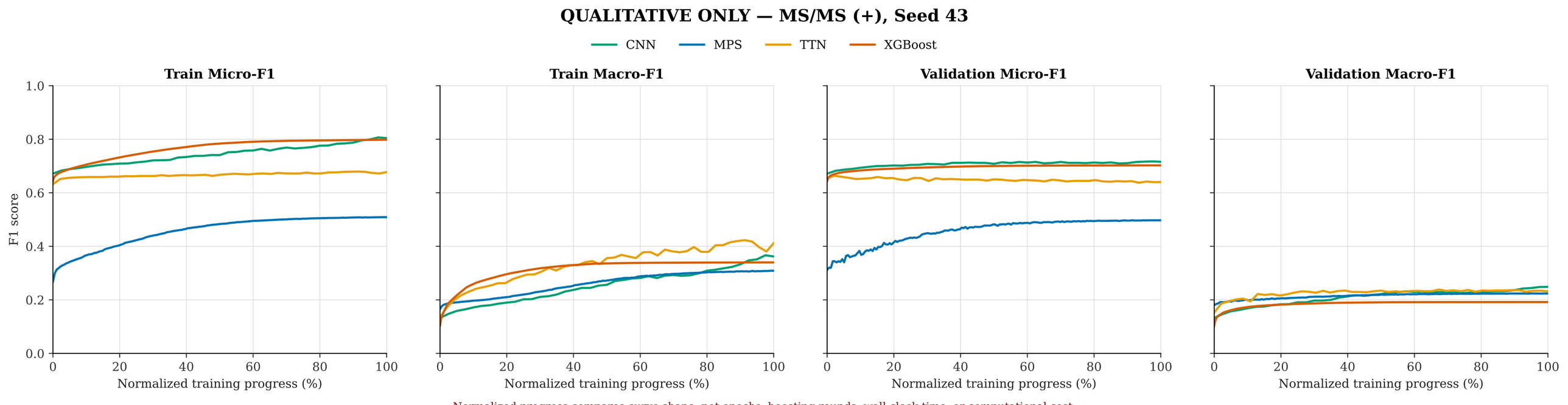

QUALITATIVE ONLY — MS/MS (+), Seed 42
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

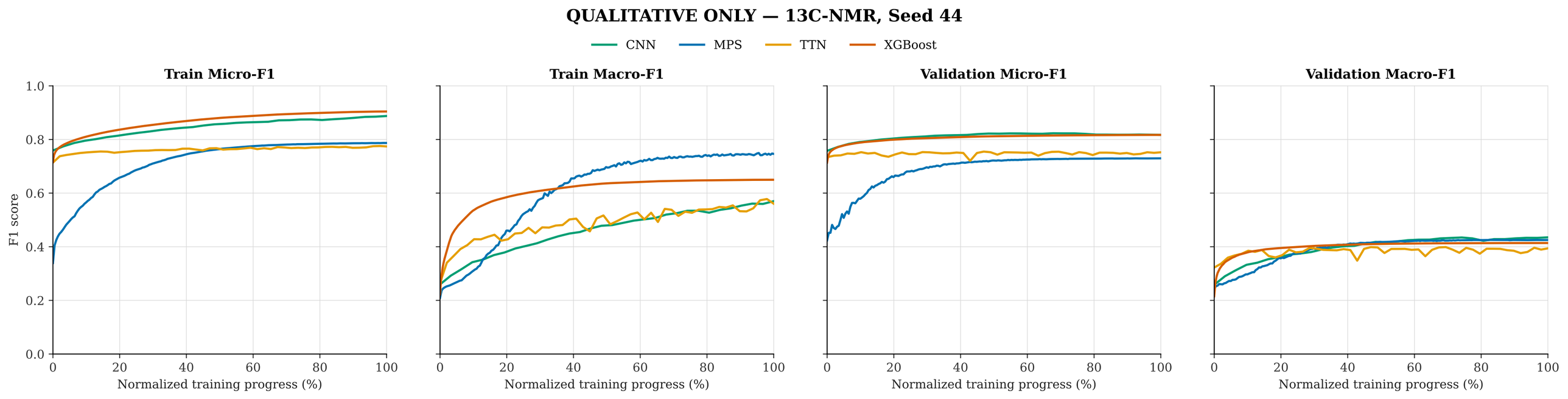

QUALITATIVE ONLY — 13C-NMR, Seed 43
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

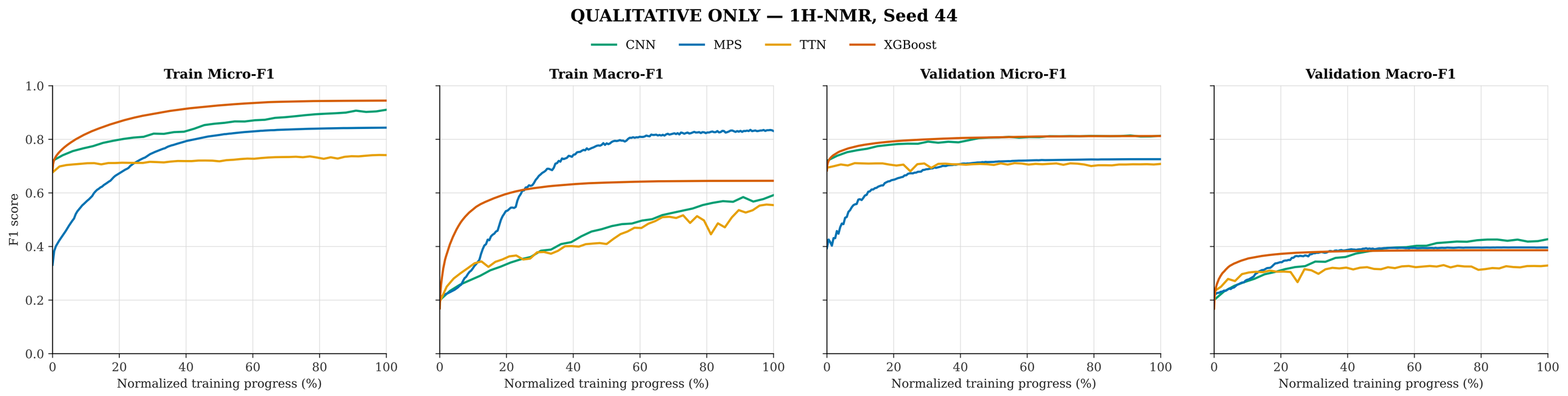

QUALITATIVE ONLY — 1H-NMR, Seed 43
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

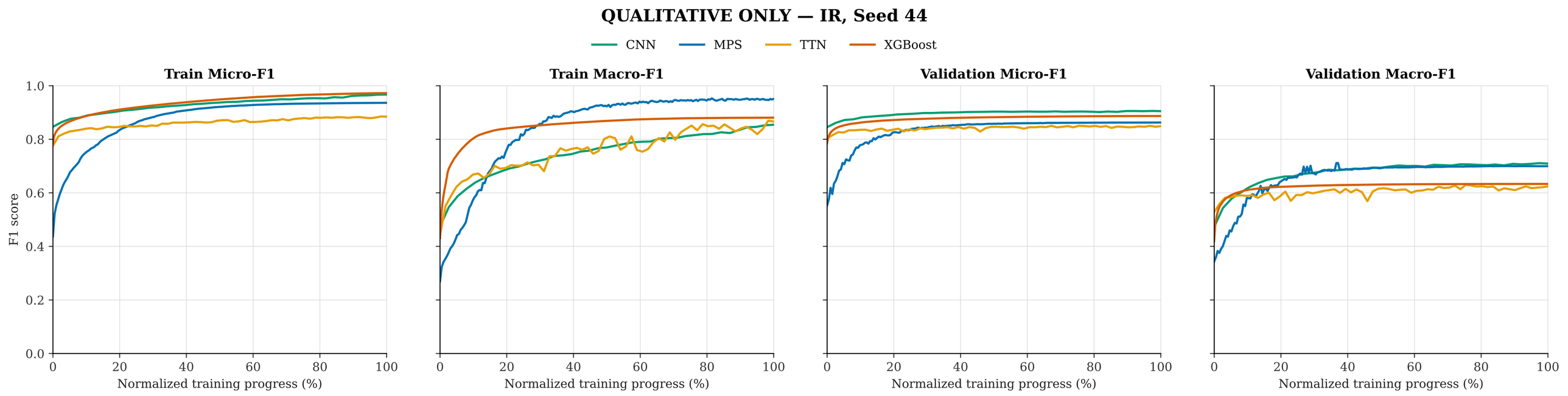

QUALITATIVE ONLY — IR, Seed 43
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

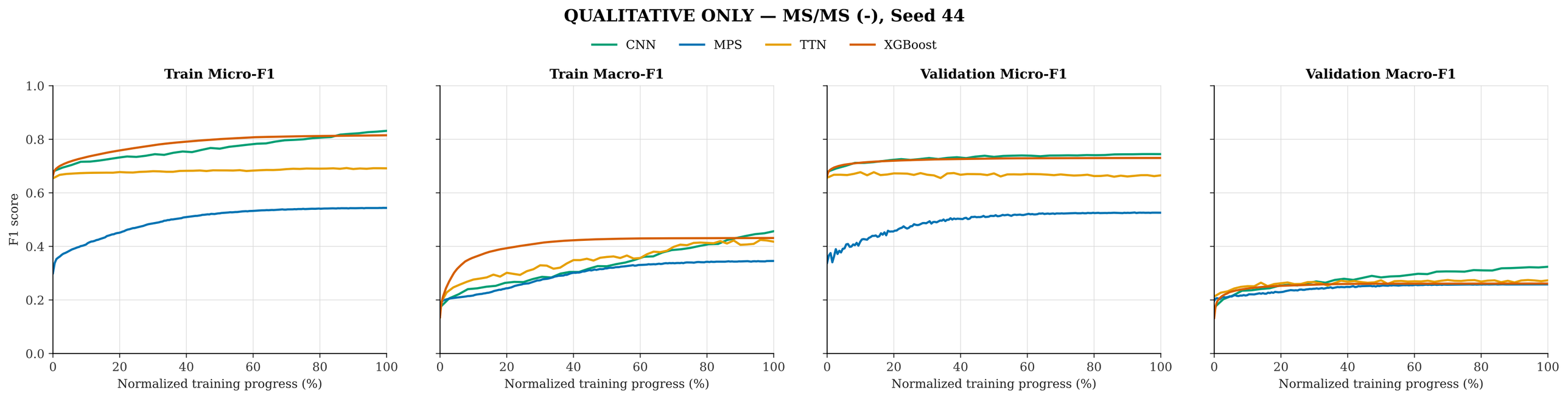

QUALITATIVE ONLY — MS/MS (-), Seed 43
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

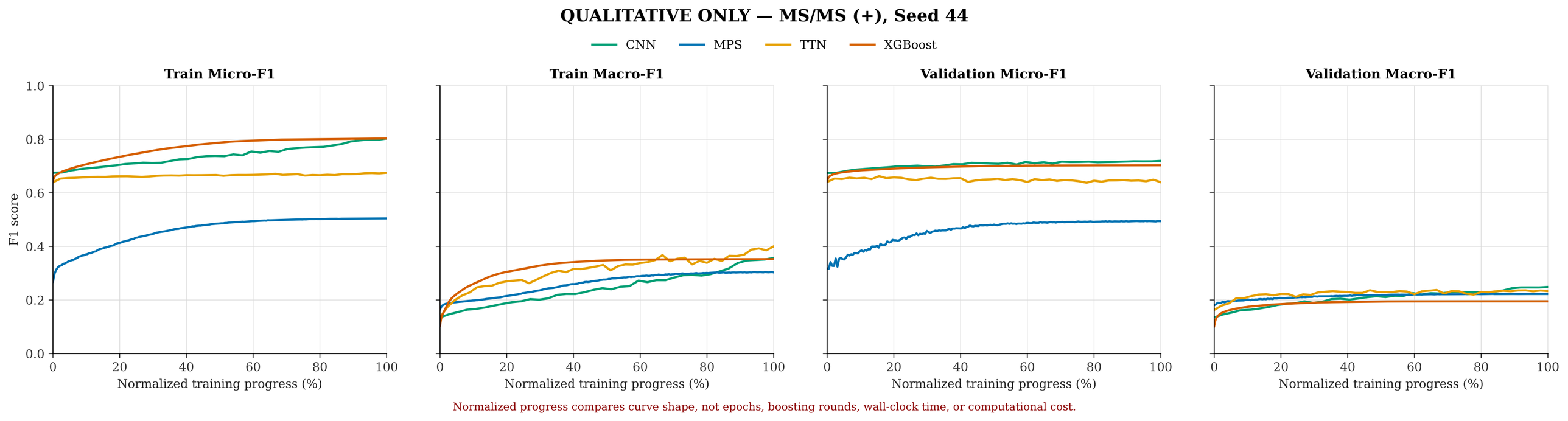

QUALITATIVE ONLY — MS/MS (+), Seed 43
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

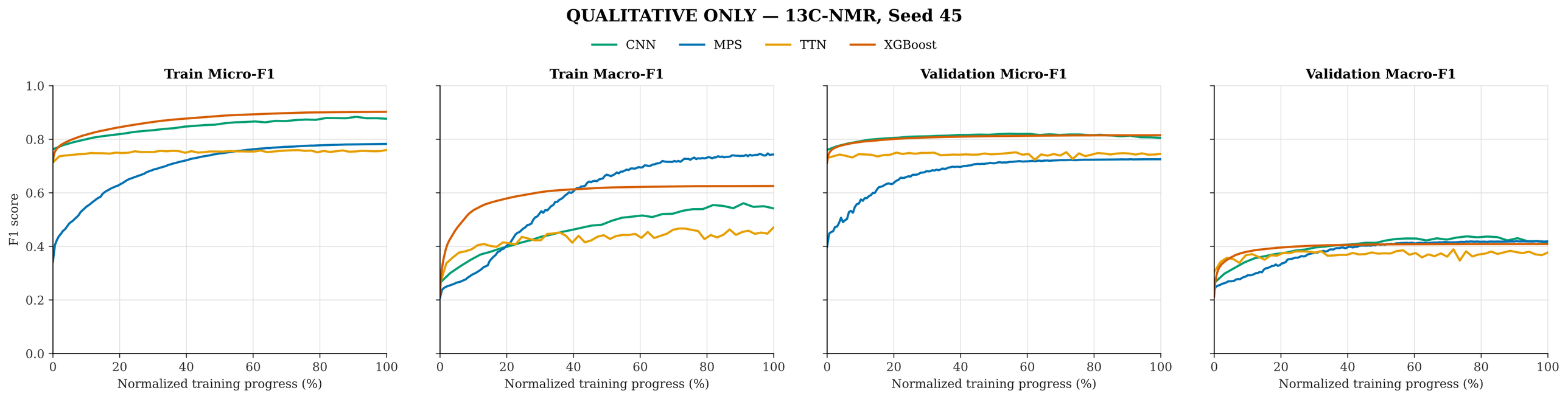

QUALITATIVE ONLY — 13C-NMR, Seed 44
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

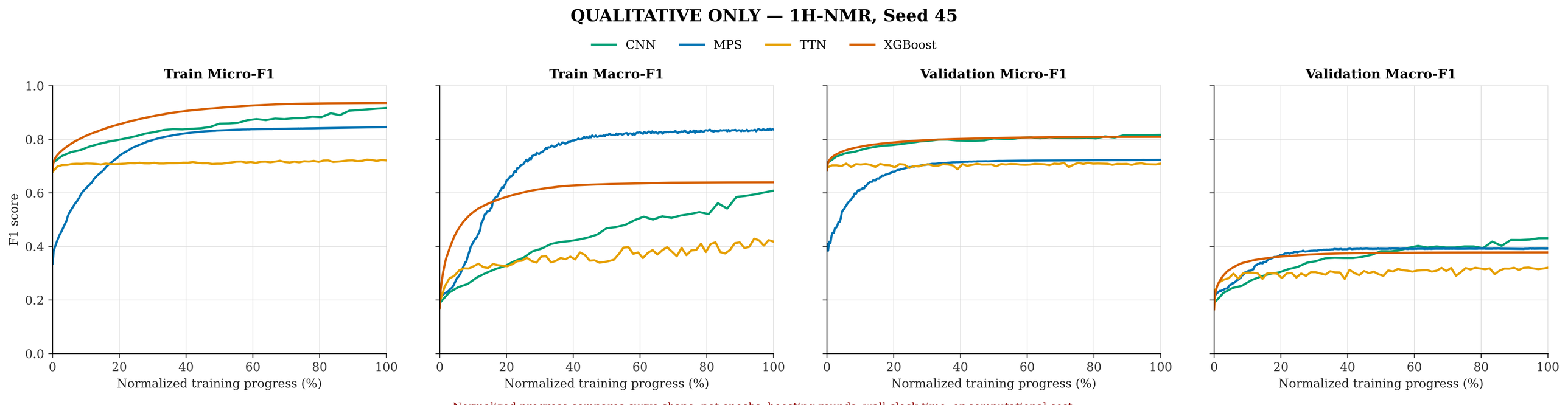

QUALITATIVE ONLY — 1H-NMR, Seed 44
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
0.0
0.2
0.4
0.6
0.8
1.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

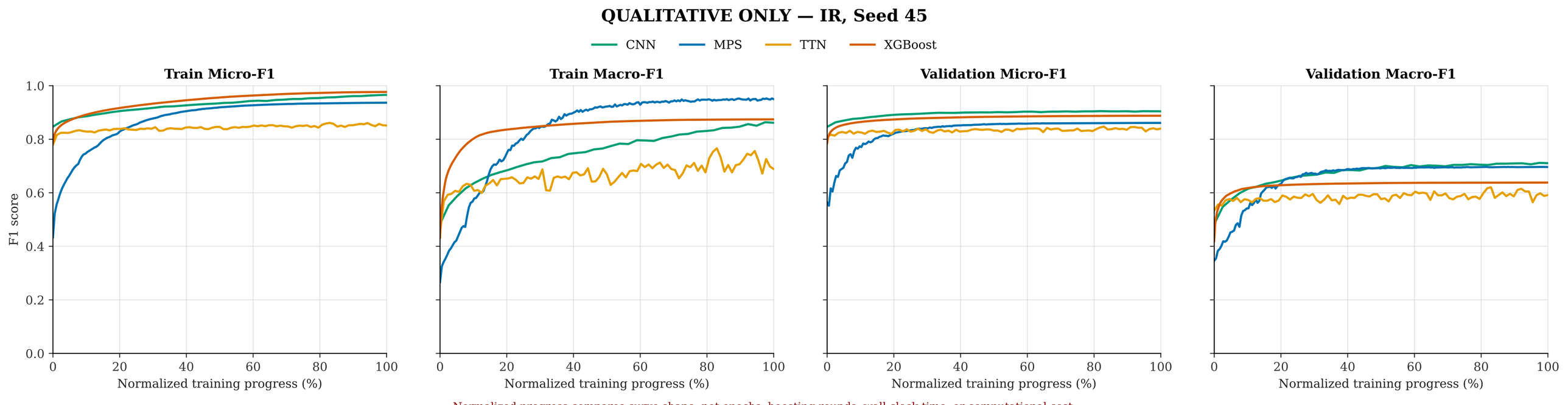

QUALITATIVE ONLY — IR, Seed 44
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

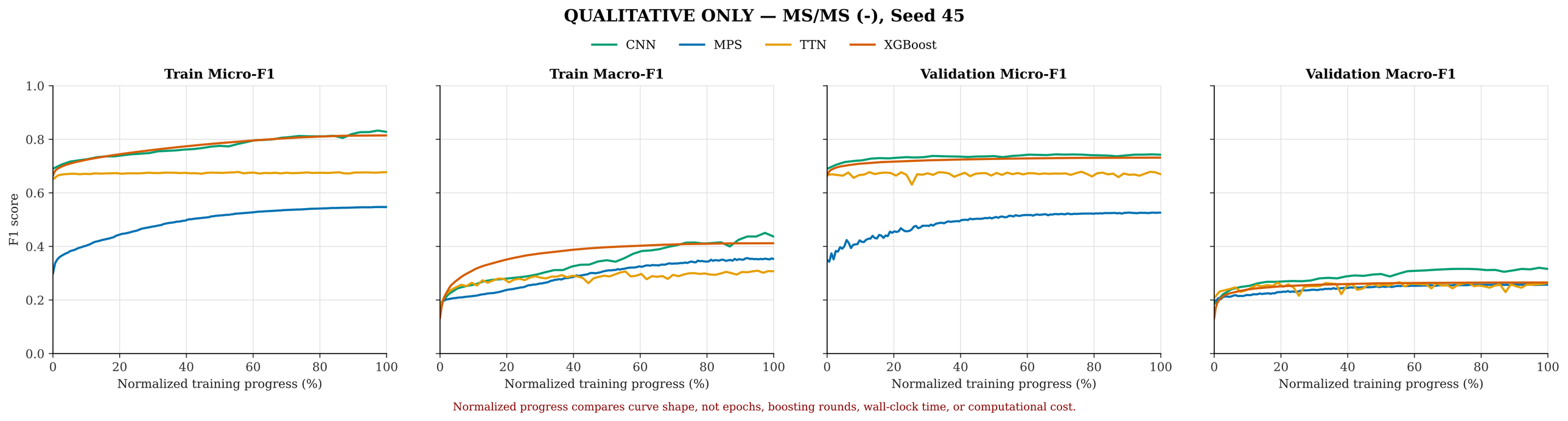

QUALITATIVE ONLY — MS/MS (-), Seed 44
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.

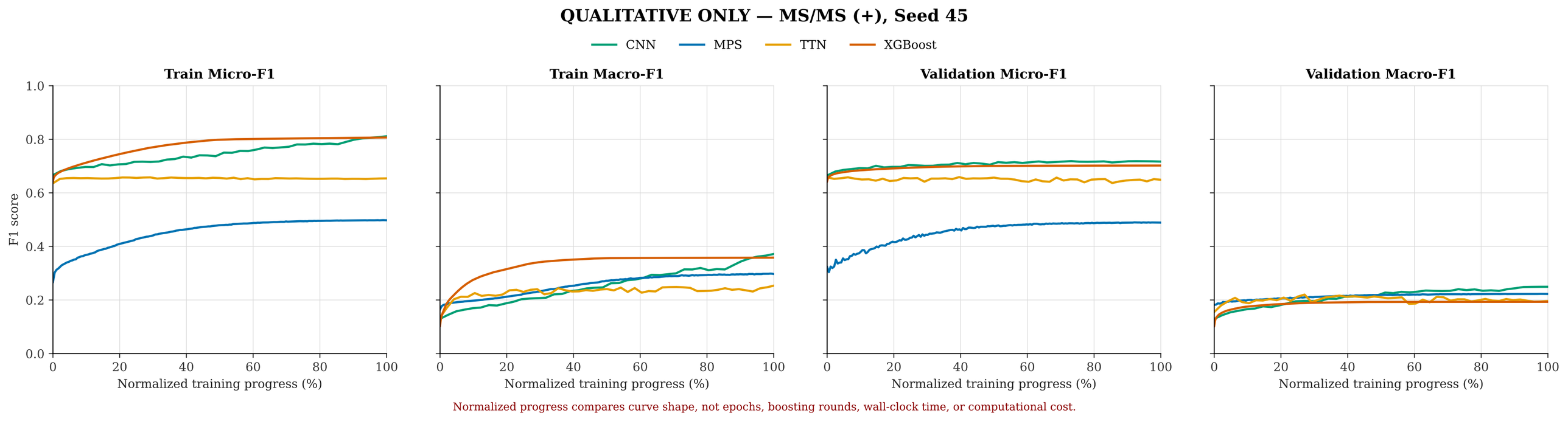

QUALITATIVE ONLY — MS/MS (+), Seed 44
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.


QUALITATIVE ONLY — 13C-NMR, Seed 45
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.


QUALITATIVE ONLY — 1H-NMR, Seed 45
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.


QUALITATIVE ONLY — IR, Seed 45
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.


QUALITATIVE ONLY — MS/MS (-), Seed 45
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.


QUALITATIVE ONLY — MS/MS (+), Seed 45
CNN
MPS
TTN
XGBoost
Train Micro-F1
Train Macro-F1
Validation Micro-F1
Validation Macro-F1
F1 score
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
Normalized training progress (%)
Normalized progress compares curve shape, not epochs, boosting rounds, wall-clock time, or computational cost.